

Human Body and Nutrition (Part - 1)

Human Body

- The human body is made up of a head, neck, torso, two legs. The average height of an adult human is about 5 to 6 feet.
- There are many systems in the human body:
- Circulatory System (heart, blood, vessels).
- Respiratory System (nose, trachea, lungs).
- Immune System (many types of protein, cells, organs, tissues).
- Skeletal System (bones).
- Excretory System (lungs, large intestine, kidneys).
- Urinary System (bladder, kidneys).
- Muscular System (muscles).
- Endocrine System (glands).
- Digestive System (mouth, oesophagus, stomach, intestines).
- Nervous System (brain, spinal cord, nerves).
- Reproductive System (male and female reproductive organs).
- Your body is made up of four main types of bones. Long bones such as the bones in your arms and legs, short bones such as the bones in your hands, feet and spine, flat bones which protect your organs provide a place for muscles to attach, and irregular bones, which are simply all the bones that are not long, short or flat.
- Your body is constantly breaking food down in the digestive system, and then using the pieces to build, repair, and grow your body.
- The job of your heart is to pump blood through your body. Even in your sleep, your heart keeps beating, or pumping blood. If it stopped, you would quickly die. Your heart is divided into two sides. The left side of your heart is filled with oxygen-rich blood, while the right side of your heart is filled with oxygen-poor blood.

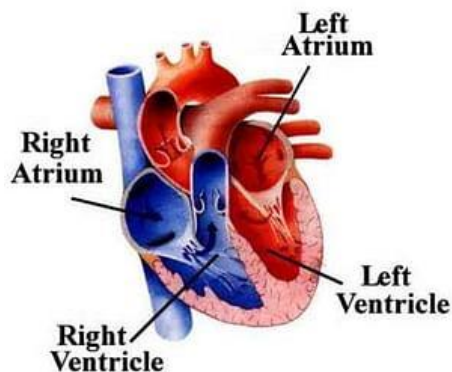
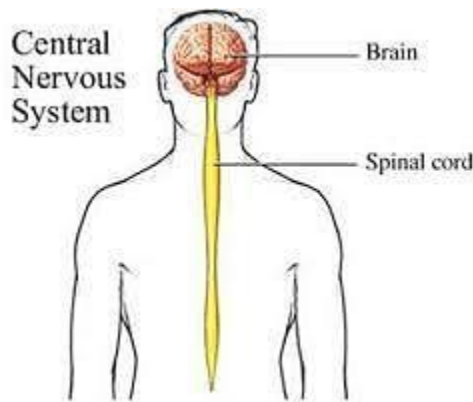


Fig: 4 Chambers of Heart

- Just like a city with hallways and roads, your body has blood vessels, arteries and veins that help move things around. Most of the cells inside your body do not move. If a cell is hungry or needs to get rid of waste, it can't simply move itself to the part of your body where it needs to go. Instead, your body must bring the food to your cells and take the waste away from them.
- Oxygen is used by your cells as it performs the functions of life. As your body uses oxygen, your cells produce another gas known as carbon dioxide. Too much carbon dioxide can be toxic, and even deadly. For this reason, it is important that your body has a way to get rid of it.
- The main organs in your respiratory system are your lungs. Your lungs are, in their simplest form, nothing more than sacs. As you breathe in, you fill these sacs with fresh oxygen-rich air. Your heart pumps blood into the walls of your lungs where it absorbs oxygen and releases carbon dioxide. As you exhale, or breathe out, you release the carbon dioxide-rich air into the space around you. With each breath you take, you are taking oxygen in, and putting carbon dioxide out.
- From your lungs, blood returns back into your heart where, it is pumped out to the rest of your body, carrying oxygen along with it.
- The central nervous system is made up of your brain and your spinal cord. It is the main control centre of your body, and the centre of thought. Your central nervous system controls most of the actions within your body.



Real-Life Example

- Children who have access to good nutritious food are more fit and active, whereas children who do not get good nutritious food and are devoid of essential vitamins and minerals suffer from various diseases and are pale, lethargic in comparison to healthy kids.



Fig: Small children deprived of food (Malnutrition)



Fig: Healthy Children

Amazing Facts

- The brain of an adult human weighs around 3 pounds (1.5kg). Although it makes up just 2% of the body's weight, it uses around 20% of its energy.
- The study of the human heart and its various disorders is known as cardiology.
- Blood makes up around 7% of the weight of a human body.
- As well as having unique fingerprints, human also have unique tongue prints.
- It takes the body around 12 hours to completely digest eaten food.
- Your sense of smell is around 10000 times more sensitive than your sense of taste.
- The toothbrush was invented more than 500 years ago by Chinese.

FOOD AND DIGESTION

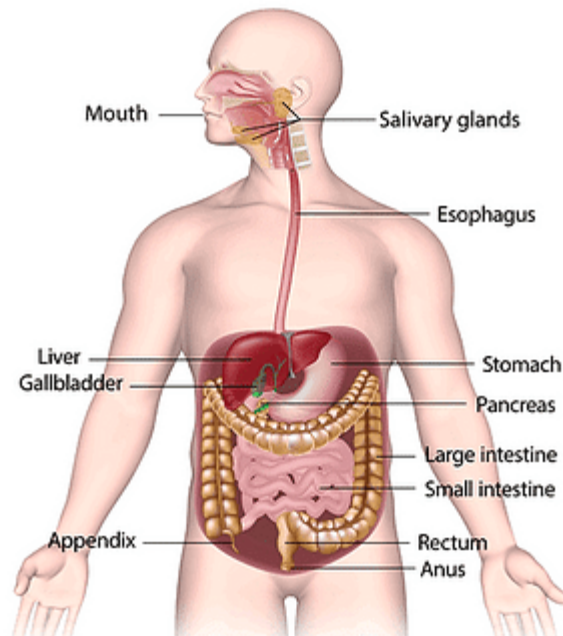


Fig: Digestive System

1. Our body must digest the food that we have eaten.
2. When we chew, food is crushed up and mixed with saliva (spit) so it becomes mushy.
3. We swallow the mush and it travels through the oesophagus, a long tube that runs from the mouth before it goes into the stomach.
4. Inside the stomach there are juices that mix the food until it looks like thick soup.
5. The food then goes to the large intestine.
6. Water from the food goes into the blood.
7. By now, the body has taken all the things it needs from the food. What is left is waste that is not needed by the body.
8. It gets stored at the end of the large intestine inside the rectum.
9. The waste must leave or the body will get sick.
10. Muscles push the waste out of the body through the anus, which is the opening in your bottom.

How long does it take for the body to digest Food?

Food can remain for 3-4 hours in the stomach, then for p about 3 more hours as it moves through the intestine It can stay in the large intestine for up to 36 hours. So depending on the type of food and the speed at which it¹ moves, it can take about 40 hours or more for the body to digest food.

- Teeth are present in mouth cavity on the upper and lower jaws. We use our teeth for chewing the food before swallowing.

- Teeth have different shapes and they are used for different purposes.
- The teeth present in front of the mouth are used for cutting. These teeth are called incisors. They are 8 in number;
- Sharp and pointed teeth are present next to incisors. These teeth are used for tearing. They are called canines. They are 4 in number.
- The flat and wide teeth are present at the back of the mouth. They are called premolars and molars. Food is crushed and grinded by these teeth. We have 8 premolars in the mouth cavity, called upper jaw and lower jaw. Each jaw has 16 teeth so every adult has 32 teeth.
- The part of tooth visible above the gum is known as crown. The part of tooth in the gum is called root which holds the tooth strongly to the gum.
- Cavities are rotten or decayed portion of the teeth that can be minor or severe, reaching even to the all tooth's core and roots.
- Cavities result from plaque build-up. As plaque accumulates, acids are released and tooth material becomes degraded. This is tooth decay. Decay is not exclusive to visible parts of one's teeth. This plaque, and its subsequent decay, can occur even at the roots and extend to the bones of the jaw.
- Taking care of your teeth helps prevent plaque which is a clear film of bacteria that sticks to your teeth.
- If you don't take care of your teeth, cavities and unhealthy gums will make your mouth very, very sore. Eating meals will be difficult.



Fig: Unhealthy gums.

SANITATION AND DISEASES

- Sanitation is the hygienic means of promoting health through prevention of human contact with the hazards of wastes as well as the treatment and proper disposal of sewage wastewater.
- Hazards can be physical, microbiological, biological or chemical agents of disease.

- Wastes that can cause health problems, include human and animal feces, solid wastes, domestic waste water (sewage, grey water), industrial wastes and agricultural wastes.



Fig: Waste in the outskirts of the city causing health problem.

- Hygienic means of prevention can be by using engineering solutions (e.g. sewerage and wastewater treatment), simple technologies (e.g. latrines, septic tanks), or even by personal hygiene practices (e.g. simple hand-washing with soap).
- Sanitation is a necessity for a healthy life.
- Lack of sanitation leads to various diseases. Some of the diseases caused are cholera, typhoid, diarrhea, etc.



Fig: Lack of Sanitation